

any number of branches, patrons, patron categories, item categories, items, currencies and other data. It also has the serials system for magazines or newspapers. It supports LDAP for authentication, and so can be easily plugged into organisations that are already running LDAP for authentication.

- **NewGenLib:** This is another ILS containing almost all the features of Koha, but with one major difference—the architecture. Koha has a two-tier architecture, whereas NewGenLib uses an n-tier architecture, with an application server in between that mediates between client machines and the database server.

Research repository

Every academic institute is known for the research papers it publishes. Every institute wants to showcase its success by this means. There are many open source software that help institutions to allow public access to these research papers:

- **DSpace:** DSpace is an institutional repository application. It has the largest community of users (including MIT, the IITs, etc) and developers worldwide, is completely customisable to fit your needs, is used by educational institutes, the government, private and commercial institutions, and can be installed out-of-the-box. It can manage and preserve all types of digital content. Moreover, it has built-in support for LDAP authentication.
- **Open Journal System:** Its basic features include editors that can configure requirements, sections, review process, etc. It can also manage online submissions and all the content, a subscription module with delayed open access options, comprehensive indexing of content, reading tools for content based on the field and editors' choice, etc. It also supports LDAP-based authentication with a plug-in.
- Other options include Fedora (www.fedora-commons.org), Greenstone, ePrints, etc.

Learning management software

These days, e-learning is part and parcel of the education system. Every institute wants academic content to be readily available to students at their convenience. At the same time, attendance management and examinations taken online save a lot of the faculty's time. There are already many good open source solutions available for this:

- **Moodle:** This is a widely used LMS across the globe, and it is supported by a big contributors' base. It has also built-in support for LDAP authentication, and a lot of plug-ins are available.
- **Sakai** is a good alternative, but is developed in Java (compared to Moodle, developed in PHP).

Group-ware

If you need to increase productivity and collaboration among

staff and students, then you need a collaboration suite that offers functionality like calendar sharing, an e-mail client, etc.

- **Group Office:** One of the best solutions, this has almost every feature anyone might need for daily office collaboration. It also comes by default with an LDAP authentication module.
- **Sogo:** A good alternative as a group-ware server, it offers a rich AJAX-based Web interface and supports multiple native clients via standard protocols such as CalDAV, CardDAV and GroupDAV. It offers users a uniform and complete interface to access their information. It supports collaboration for users of Mozilla Thunderbird/Lightning, Microsoft Outlook, Apple iCal/iPhone and BlackBerry, through the use of a modem platform.

Content management systems

Every academic institute needs content management, to display various information related to academic programs, admission procedures, etc, in an easily manageable way. There are many open source solutions available, with similar functionality, but based on your needs, you can decide which suits you the best. All the following solutions offer authentication against LDAP:

- Drupal
- WordPress
- Liferay
- Joomla

Student management software (ERP)

Academic ERP can be used to decrease the administrative workload. It includes functionality like student registrations, fees collection, time tables, parents-teacher interactions, etc. Listed below are a few such solutions:

- **Fedena:** Based on Ruby On Rails, this is one of the best academic open source ERP implementations till date. It also offers features for library management, but these are limited, compared to the requirements of big academic institutions.
- **OpenSIS:** A PHP-based implementation of academic ERP, this is very limited compared to Fedena.

Question and answer forum sections

Every institution needs some type of Q&A platform, which can act as an FAQs page, so that admission-related and other student and staff queries can be solved, and archived for future use. Here's an open source solution for this:

- **QSQA:** It supports LDAP authentication, and is one of the best open source Q&A solutions till date. 

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